

2 How Maps Are Made

Every map starts as a set of measurements. Surveyors walk the land with instruments that record angles and distances, satellites send down photographs taken from hundreds of kilometres above, and volunteers trace roads from those photographs on their home computers. None of these sources is complete on its own, so mapmakers combine them, check them against each other and decide what the reader really needs to see.

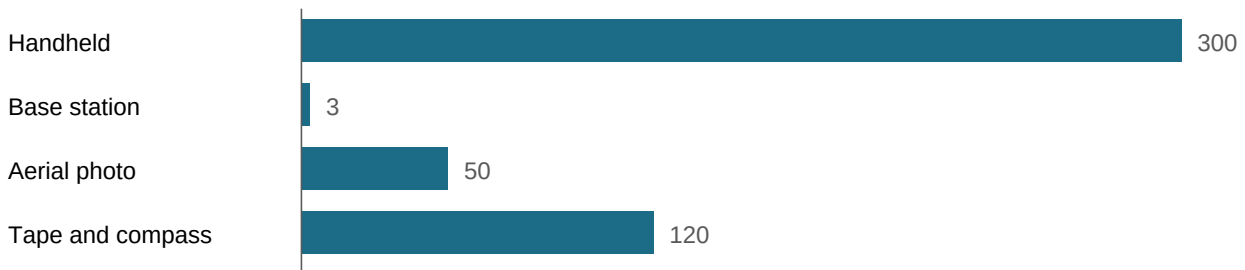
This chapter follows a map from the first measurement to the printed page. We look at how positions are measured, why a flat map can never show the round Earth without distortion, and how symbols, colours and labels turn raw data into something a person can read at a glance.

2.1 Measuring the ground

A position on Earth is written as a pair of numbers: latitude, the angle north or south of the equator, and longitude, the angle east or west of a chosen line. Modern receivers find both by timing signals from several satellites. The table of methods below shows how accurate each approach is.

- Handheld receiver: a few metres, good for walking routes.
- Survey receiver with a base station: a few centimetres.
- Aerial photographs: about half a metre, fast for large areas.

Figure 2.1 Typical error of each method (centimetres)



Note

A map is always a choice. Leaving out small paths makes a city map easier to read, but a hiking map needs exactly those paths.

2.2 From a round Earth to a flat page

Peel an orange and try to press the peel flat: it tears or stretches. A map projection is a rule for doing the same with the Earth's surface, and every rule has to give something up. Some keep the shapes of countries, some keep their areas, and some keep the directions a sailor needs.